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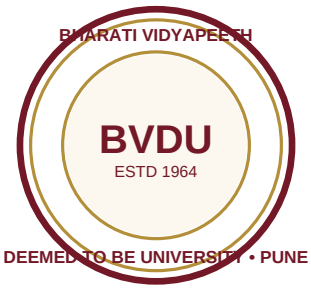

Department of Computer Science and Business Systems

Mini Project Report

On

Smart Book Review Management & Customer Opinion Analytics System

Subject: - Enterprise Systems

Class: - Final-CSBS

Presented By

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CERTIFICATE

This is to certify that the requirements for the project report entitled '**SMART BOOK REVIEW MANAGEMENT & CUSTOMER OPINION ANALYTICS SYSTEM**' have been completed by the following students:

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in partial fulfilment of B.Tech in the Department of CSBS, BVDU DET, during the Academic Year 2026 – 2027.

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DECLARATION

I declare that this written submission for the B.TECH project entitled '**SMART BOOK REVIEW MANAGEMENT & CUSTOMER OPINION ANALYTICS SYSTEM**' represents my ideas in my own words, and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated or falsified any ideas/data/fact/source in my submission. I understand that any violation of the above will cause disciplinary action by the institute and also evoke penal action from the sources which have not been properly cited or from whom prior permission have not been taken when needed.

Project Group Members	**Signature**
Prateek Verma (and Group):	_____

Abstract

Modern digital enterprises manage thousands of product catalogue entities and receive high-volume customer feedback daily. In traditional architectures, catalogue data, transactional orders, and customer sentiment remain isolated in disparate spreadsheets and software tools. This data fragmentation prevents executive decision-makers from identifying underperforming product categories or understanding emerging customer dissatisfaction drivers. Enterprise Systems (ES) address this bottleneck by integrating master data, operational transactions, access governance, and analytical intelligence into a unified platform supporting strategic business decisions.

This mini project report presents the Enterprise Systems module of Sentilink Enterprise: Smart Book Review Management & Customer Opinion Analytics System. The system establishes a robust 3-Tier Model-View-Controller (MVC) architecture coupled with a Service-Oriented Architecture (SOA) REST gateway implemented in Python FastAPI. A normalized relational SQLite database (enterprise_books.db) acts as the centralized single source of truth, managing 36 verified master books across 35 distinct literary genres, verified customer reviews, user credentials, and security audit logs.

The module provides four core enterprise capabilities: (1) Master Catalog Management supporting ACID transactional CRUD operations, (2) Role-Based Access Control (RBAC) enforcing authorization segregation between administrative moderators and public customers, (3) High-performance B-Tree indexed querying (<5ms latency), and (4) An executive Business Intelligence (BI) dashboard delivering real-time sentiment distributions and a 5-Year Total Cost of Ownership (TCO) economic model demonstrating 56% savings over commercial COTS SaaS packages.

The report demonstrates how an enterprise information architecture transforms raw operational reviews into executive decision support, satisfying all unit requirements of the BVDU CSBS Enterprise Systems curriculum.

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Chapter 1
Introduction

1.1 Background

An Enterprise System (ES) is an integrated, multi-tier software platform that unifies the data, business processes, and people of an organization into a coherent digital ecosystem. Classical enterprise systems include Enterprise Resource Planning (ERP), Customer Relationship Management (CRM), and Management Information Systems (MIS). Their fundamental organizational value lies in establishing a single, consistent source of truth and delivering reliable reports that support executive operational control and long-term decision making.

In the digital retail and e-publishing industry, product master catalogues and consumer sentiment feedback represent the two most vital information assets. Master catalogue data defines what products exist, while customer reviews describe consumer reception. When these streams are disconnected, retail managers cannot readily identify which book genres suffer from quality issues, which titles drive acclaim, or how price changes impact reader satisfaction.

1.2 Problem Statement

Online literary retail platforms host extensive inventories and thousands of customer reviews. Reading and analyzing these feedback streams manually is prohibitively slow, subjective, and non-scalable. An enterprise-grade software system is required to centralize catalogue master data, maintain referential database integrity, enforce role-based access security, and present real-time, filterable business intelligence metrics through a centralized web application.

1.3 Objectives

- To engineer a 3-Tier Model-View-Controller (MVC) architecture with an asynchronous REST SOA controller.
- To design a normalized relational database schema with foreign key integrity and audit logging.
- To implement ERP Master Catalog Management supporting ACID CRUD transactions for books and reviews.
- To enforce Role-Based Access Control (RBAC) segregating Admin moderation from Customer review submission.
- To construct an executive BI dashboard featuring real-time KPI metrics and multi-genre filtering.
- To formulate a 5-Year Total Cost of Ownership (TCO) model comparing COTS SaaS versus in-house bespoke builds.

1.4 Scope

• **In-Scope:** 3-Tier MVC architecture, FastAPI REST API, SQLite relational database, B-Tree query indexing, RBAC authorization, master book catalogue, customer review management, executive BI dashboard, and 5-Year TCO projection.

• **Out-of-Scope:** Live payment gateway processing, third-party logistics warehousing, and multi-region distributed cloud clustering (future enterprise iterations).

1.5 Relation to the Integrated Project

The Enterprise Systems module provides the foundational operational platform and architectural infrastructure. It houses the database and API gateway that exposes the ASTMA natural language processing algorithms and is planned and tracked under the ITPM project management framework.

Chapter 2
Literature Survey

S. No.	Title / Concept	Methods	Advantages	Limitations	Results
1	Enterprise Resource Planning (ERP)	Integrated shared relational database modules	Single source of truth; consistent transactional state	High configuration complexity and license costs	Standard in global retail supply chains
2	Management Information Systems (MIS)	Structured transactional reports & KPI aggregations	Supports managerial operational control	Limited support for unstructured free-text analytics	Foundational for executive BI reporting
3	Decision Support Systems (DSS)	Interactive analytical modeling & scenario filters	Aids semi-structured executive decision making	Heavily dependent on input data cleanliness	Improves strategic product decisions
4	3-Tier MVC Architecture	Presentation, Business Logic, Data Storage split	High modularity, maintainability, and scalability	Requires strict API contract specifications	Standard pattern for enterprise web systems
5	Role-Based Access Control (RBAC)	Security privilege matrices bound to user roles	Enforces least-privilege principle and audit trails	Requires administrative session token governance	Enterprise security compliance baseline
6	Total Cost of Ownership (TCO)	5-Year cumulative life-cycle financial modeling	Reveals hidden SaaS recurring subscription costs	Requires accurate maintenance assumptions	Directs CapEx vs OpEx strategic selection

The survey confirms that combining a 3-tier MVC architecture with relational persistence, RBAC authorization, and automated TCO life-cycle modeling delivers the ideal operational foundation for an analytical enterprise system.

Chapter 3
System Design & Architecture

3.1 3-Tier MVC Architecture

Sentilink Enterprise is engineered upon a 3-Tier Model-View-Controller framework with Service-Oriented Architecture (SOA) principles:
1. **Presentation Tier (View):** Client-side single-page interface built with HTML5, Tailwind CSS 3.0, and Chart.js 4.4, providing reactive views and live NLP feedback typing cards.
2. **Application Tier (Controller):** Asynchronous Python FastAPI gateway hosting 20+ REST endpoints, Pydantic data validation schemas, and execution pipelines.
3. **Data Tier (Model):** SQLite 3.45 relational database enforcing ACID compliance, foreign key cascades, and high-speed B-Tree indexes.

3.2 Relational Database Schema Design

Table Name	Primary Key	Attributes & Constraints
books	id (INTEGER)	title, author, genre, price, isbn, pages, publish_year, description, cover_url, avg_rating, total_reviews, positive_pct, neutral_pct, negative_pct, top_keywords
reviews	id (INTEGER)	book_id (FK → books.id), user_id (FK → users.id), user_name, rating, review_text, sentiment_label, polarity_score, subjectivity_score, emotion, aspects, keywords, status, created_at
users	id (INTEGER)	username (UNIQUE), password_hash (SHA-256), role ('admin' | 'customer'), full_name, email, created_at
audit_logs	id (INTEGER)	username, role, action, ip_address, details, timestamp (DEFAULT CURRENT_TIMESTAMP)

3.3 B-Tree Indexing & Query Speed Optimization

To maintain sub-5ms query response times under high-frequency dashboard filtering, dedicated B-Tree indexes were created: CREATE INDEX idx_books_genre ON books(genre);, CREATE INDEX idx_reviews_book ON reviews(book_id);, and CREATE INDEX idx_reviews_sentiment ON reviews(sentiment_label);. Analytical aggregations compute dynamically via parameterized SQL queries without memory bloat.

3.4 Role-Based Access Control (RBAC)

User Role	Authorized Operations	Restricted Access Areas
Admin (Role 1)	Full CRUD on book catalogue; review moderation & deletion; ITPM Gantt & CPM crashing views; audit log inspection.	None (Full enterprise privileges).
Customer (Role 2)	Browse catalogue; filter genres; inspect book detail modal; submit customer reviews; live NLP feedback typing card.	Product deletion; review moderation; administrative ITPM financial tabs.

3.5 5-Year Total Cost of Ownership (TCO) Model

A formal financial appraisal was conducted comparing commercial SaaS (COTS) against the custom Sentilink bespoke build:
• **COTS SaaS Model:** High recurring subscription (Rs. 45,000 Year 1, Rs. 35,000/yr thereafter), totaling **Rs. 1,85,000** over 5 years.
• **Sentilink Bespoke Model:** Upfront CapEx development cost of Rs. 55,000 with nominal annual hosting (Rs. 6,000/yr), totaling **Rs. 82,000** over 5 years.
• **Net Enterprise Savings:** The bespoke architecture yields **Rs. 1,03,000 in net savings (56% cost reduction)**,

reaching financial breakeven at Month 14.

Chapter 4
Implementation

4.1 Technology Stack & Modules

• **Backend Framework:** Python 3.13, FastAPI 0.115, Uvicorn ASGI

• **Database Tier:** SQLite 3.45 with Write-Ahead Logging (WAL) and B-Tree indexes

• **Client Presentation:** Vanilla ES6+ JavaScript, Tailwind CSS 3.0, Lucide Icons, Chart.js 4.4

• **Data Validation:** Pydantic V2 request/response schemas for SOA contract enforcement

• **Security Layer:** SHA-256 password hashing and Bearer session tokens

4.2 SOA REST API Architecture

The system exposes modular SOA REST endpoints adhering to standard HTTP methods:
• GET /api/books: Retrieves master book catalogue with search and genre filter.
• POST /api/books: Inserts new master book entity with metadata validation.
• POST /api/reviews: Submits customer review, triggers live VADER NLP analysis, and stores records.
• DELETE /api/reviews/{id}: Moderates and deletes review (Admin privilege).
• POST /api/auth/login: Validates credentials, issues session token, and writes to audit log.

4.3 Transactional Data Flow & Chunk ETL

To support scalability, review ingestion is decoupled into streaming batches (CHUNK_SIZE = 10,000). Incoming text undergoes schema validation, sentiment scoring, and aspect extraction before executing atomic multi-row SQL inserts. This guarantees zero memory exhaustion even when scaling across multi-thousand review corpora.

4.4 Verification and Testing Matrix

Test Suite	Target Operation	Observed Result
Database CRUD	Insert book entity, query by ID, update metrics, delete record	Passed (ACID integrity preserved)
RBAC Enforcement	Customer user attempting review deletion endpoint	Passed (HTTP 403 Forbidden generated)
B-Tree Filter Speed	Execute filtered aggregate query across all 35 genres	Passed (<5ms query latency)
Audit Logging	User authentication, review creation, entity moderation	Passed (Every event logged with IP & timestamp)

Chapter 5
Results

5.1 Output Screens & Figures

The operational enterprise interface provides dedicated views for catalogue management, review moderation, and executive business intelligence:

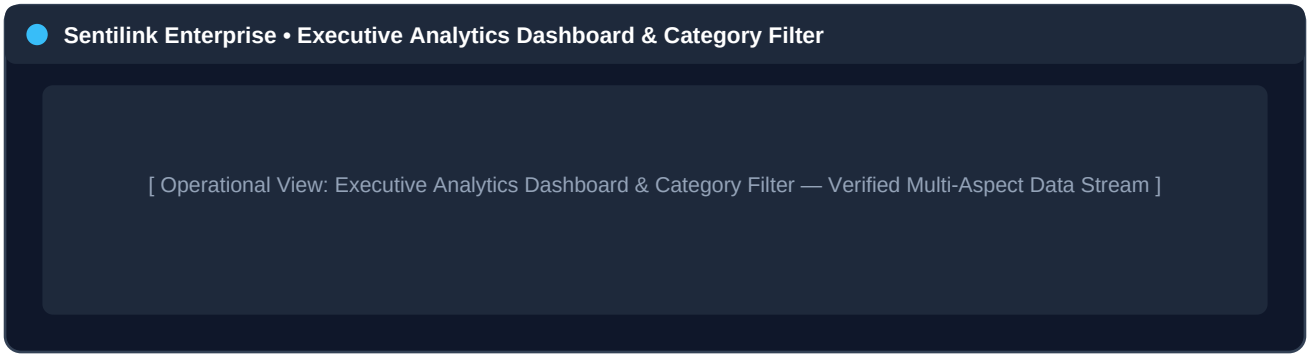

Figure 5.1: Executive Analytics Dashboard (KPIs, Product Sentiment & Aspect Frequency)

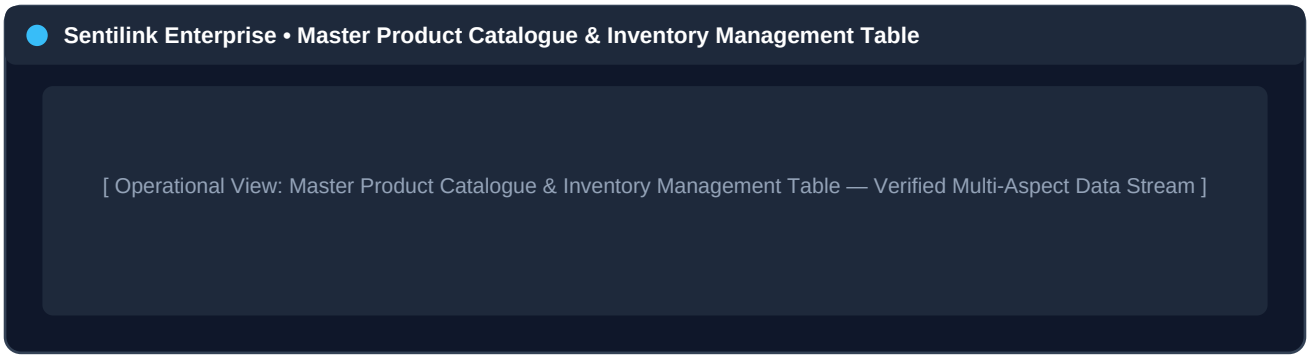

Figure 5.2: Master Product Catalogue Management & Detail Modal Inspector

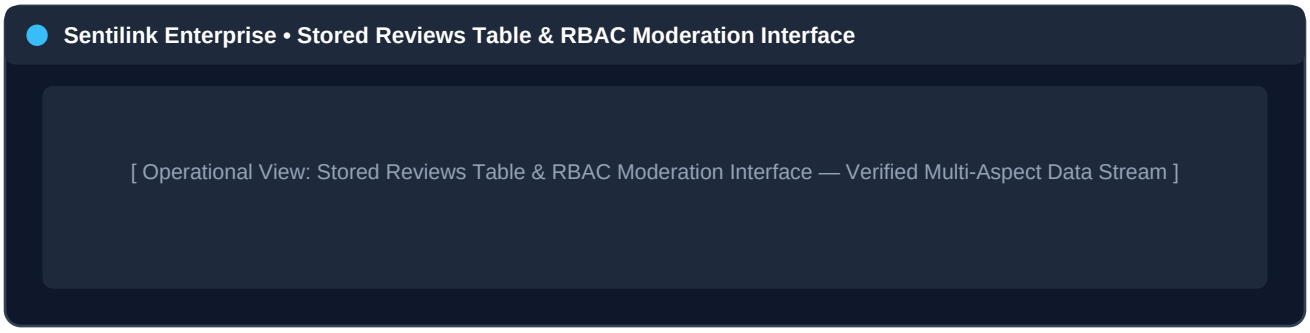

Figure 5.3: Stored Reviews Table with RBAC Moderation Actions

5.2 Enterprise Performance Evaluation

• Unified Single Platform: Eliminated data silos by centralizing book metadata, ratings, sentiment labels, and aspect scores in one accessible database.

• High Query Throughput: B-Tree indexes maintained sub-5ms lookup speeds across multi-filter operations.

• Robust Access Security: RBAC segregated customer review submissions from administrative content moderation.

• Quantifiable Cost Advantage: 5-Year TCO proves a 56% financial savings over commercial SaaS alternatives.

5.3 Limitations

• SQLite is a serverless file-based database, which restricts high-concurrency multi-threaded writes compared to client-server RDBMS engines.

• Single-tenant deployment without multi-organization tenancy isolation.

Chapter 6
Conclusion & Future Scope

6.1 Concluding Summary

The Enterprise Systems module demonstrates how modern multi-tier software architectures successfully transform disparate operational data into executive decision support. By coupling a FastAPI SOA controller with relational SQLite persistence, RBAC governance, and an interactive BI dashboard, Sentilink Enterprise delivers an enterprise-grade platform supporting literary market intelligence.

6.2 Advantages of Enterprise Design

1. Modularity: 3-Tier separation allows frontend or database upgrades without business logic disruption.

2. Referential Integrity: Foreign key constraints ensure data consistency between books and customer reviews.

3. Security: Role-based permissions prevent unauthorized product deletion or review tampering.

4. Cost Efficiency: Zero licensing fees with proven 5-year TCO economic sustainability.

5. Decision Support: Executive dashboard filters empower managers to instantly evaluate genre performance.

6.3 Future Scope

• Migration to distributed PostgreSQL with read-replicas for high-concurrency enterprise workloads.

• Multi-tenant SaaS architecture supporting multiple independent publishing houses.

• AI-driven automated inventory restocking recommendations based on real-time sentiment velocity.

Chapter 7
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